Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Pecel Lele
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, 5, 8, 9, 10
Project / Solution Name	Adept
Name (Leader)	David Kurniawan
Email & Phone (Leader)	davidk.academic@gmail.com & +6281229700811
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

As undergraduate students preparing to enter Malaysia's workforce, our team is tackling a question that directly concerns our future: how can we identify and develop skills that remain relevant as AI changes the work employers need? Decisions about electives, projects, internships and additional training require time and money, yet their connection to future career requirements can be unclear.

The urgency extends beyond graduation. Hays Malaysia reported in March 2025 that 56% of working professionals used AI tools at work, while 26% had received training to use them. These findings highlight the importance of preparing people to adapt alongside technological change. (Hays Malaysia, 2025)

Choosing a useful learning opportunity requires connecting existing skills, changing job requirements, course prerequisites, costs and available study time. Students must determine which gaps deserve attention and how individual courses fit into a realistic career direction. Working adults and career switchers face similar decisions while balancing employment and other responsibilities.

Poorly informed choices can consume limited learning budgets and delay preparation for suitable roles. People with less access to professional guidance face additional barriers, while employers and career advisors must identify relevant training across different individuals.

Our team therefore focuses on making the connection between skills, career options and learning opportunities clearer and more actionable. This challenge aligns with SDG 4's emphasis on relevant, accessible education and SDG 8's focus on productive employment and adaptation to technological change.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Adept helps individuals and organisations turn existing skills into practical reskilling decisions. Users complete a questionnaire or upload a synthetic CV, confirm extracted skills and proficiency, and set interests, available study time and budget.

- Dynamic Trend Ingestion: A batch pipeline combines arXiv research-attention signals with a BLS-derived labour baseline. The Future Skills Radar presents dated snapshots and moderate/fast scenarios as estimates, rather than real-time Malaysian employment forecasts.
- Explainable Matching & Gap Analysis: The default score combines skill fit (45%), trend outlook (25%) and interest fit (30%), with optional SDG weighting. Only interest fit uses Jaccard similarity. Skill fit compares proficiency with each role's requirements. Skills are Have when the required level is met, Partial when present below that level, and Missing when absent.
- Grounded Reskilling Pathways: Deterministic rules select catalogue courses and order learning by prerequisites, skill gaps, time and budget. Adi optionally uses GLM-5.3, with Gemini failover, to select validated application-authored answer blocks; model-written prose is not displayed.
- Meaningful SDG Alignment: Team-curated role and skill mappings connect career and learning choices to SDG targets. Contribution scores describe alignment, not measured social outcomes or UN endorsement.

Organisation users can inspect shared skill gaps, create training programmes, allocate funding and review completion evidence. Public job postings and dated certification fees connect learning to practical opportunities.

Technical Foundation (SDC Phase 1): GenAI integration, relational database design and cloud deployment connect through Next.js, private FastAPI, Replit integrated PostgreSQL/pgvector, RunPod MiniLM and guarded model-provider calls. Sponsor-provided Replit Core hosts the public prototype; RunPod provides US$25 in credits.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Adept operates on a modular architecture driven by a single backend that supports two distinct user experiences: an Individual Mode for job seekers and career switchers, and an Organisation Mode for HR teams and academic advisors. This system seamlessly integrates user skill assessments, AI-driven market trends, transparent role matching, custom learning roadmaps, and UN SDG alignments.

Core Technologies, Frameworks, and Databases:

- **Frontend UI:** Next.js 16.4.2, React 19.2.9, TypeScript ~5.9.3, Tailwind CSS 4.1.18, lucide-react icons, and Three.js 0.180.0 (Adi's interactive 3D mascot). Localised in English, Malay and Chinese via next-intl 4.14.9.
- **API proxy layer:** Next.js route handlers forward same-origin API requests to private FastAPI on loopback. Signed session cookies and a trusted internal proxy boundary connect the public interface to the backend.
- **Backend API:** Python ≥3.11 (python:3.11-slim image), FastAPI ≥0.161.1, Pydantic v2 with pydantic-settings, Uvicorn, HTTPX, local sessions, and python-multipart for CV uploads.
- **API integration:** OpenAPI contracts generate typed TypeScript bindings using openapi-typescript 7.9.1 and openapi-fetch 0.14.1.
- **Database:** Replit integrated PostgreSQL provides the deployed application database, accessed through psycopg 3 and psycopg-pool. It stores accounts, profiles, catalogue records, plans, progress, organisation data and vector indexes. Local development and CI use PostgreSQL 16 with pgvector.
- **AI & data processing:** all-MiniLM-L6-v2 (pinned revision) runs on a RunPod serverless worker (sentence-transformers 6.2.0, Transformers 5.20.2, PyTorch 2.14.1, RunPod SDK 1.12.0) to generate 384-dimensional embeddings offline for skills and

research signals. Optional Gemini Embedding 001 (768-dimensional) supports catalogue and skill similarity, and short phrases are redacted before sending.

- **Infrastructure & QA:** Node.js 24 with pnpm 10.29.3. Docker Compose locally, GitHub Actions CI with scheduled trend refreshes, and pytest, Ruff, ESLint 9.39.5 and strict TypeScript checks.

Public Data and API Integrations: The platform ingests arXiv cs.AI feeds, BLS 2023–2033 projections (a US labour baseline, not a Malaysian vacancy forecast), the O*NET 31.0 crosswalk, and the UN SDG Goal/Target APIs. It also uses public job-posting feeds (Greenhouse, Lever, Ashby, Workday, Jibe, Himalayas, Jobicy, Remote OK, Remotive; the committed 23 Oct 2026 extract covers nine of these), certification and exam-pricing pages (Microsoft, AWS, Google Cloud, IIBA, IAPP), and course catalogues (Microsoft Learn, Coursera, plus Malaysian providers such as FMM, MAICSA, UKM, UTM, UNIMAS PROCEL, IBFIM and AITraining2U). The Bank Negara Malaysia API converts prices to MYR. Optional keyed job APIs (Careerjet, Jooble, Web3.career) have connectors but are currently skipped. Every recommendation carries attribution metadata: sources, snapshot_date, scenario and is_estimate.

Deployment Strategy & GenAI Usage: Adept is publicly deployed at https://adept-my.replit.app using sponsor-provided Replit Core and Replit integrated PostgreSQL. A Python supervisor runs one public Next.js server and a private FastAPI service. Local Docker Compose provides PostgreSQL only. Adi uses configured GLM-5.3 inference on RunPod, with optional Gemini failover when separately configured and authenticated. The model selects validated application-authored answer-block IDs; displayed answers come from those blocks. Provider or validation failures use a deterministic guide.

How Workshop Elements Connect:

1. **Offline Signal Layer:** Public data (arXiv research volume, BLS economic baselines, UN SDG targets, Microsoft Learn courses) is periodically fetched, scored offline using the MiniLM encoder, and committed to PostgreSQL as validated snapshots.
2. **Deterministic Processing Engine:** When users submit profiles, FastAPI evaluates skill gaps, projects future demand under moderate/fast scenarios, topologically sorts

prerequisite courses into time-bound pathways, and computes SDG alignments deterministically.

3. **Guardrails & Attribution:** Catalogue recommendations resolve to existing database IDs. Adi selects validated answer blocks instead of composing new recommendations. Relevant outputs carry source, snapshot, scenario and estimate metadata. Input filtering reduces exposure of personal information but does not guarantee complete anonymisation.
4. **Dual Frontend Views:** Next.js serves both Individual Mode (personal career reskilling) and Organisation Mode (workforce skills-gap heatmaps and group plans) from this single unified backend.

SDC Sponsor Utilization (GenAI API & Cloud Platforms)

- **Domain Sponsor (Aiversecity):** Adept addresses the domain's future-ready workforce challenge through explainable skills matching and reskilling support. This is domain alignment, rather than a claim that an Aiversecity API was used.
- **Platform Sponsors (Runpod, Replit):**
    - Runpod: US$25 sponsor-provided credits support the MiniLM Serverless embedding worker and the separate GLM-5.3 inference endpoint used by Adi.
    - Replit: The sponsor-provided Core subscription hosts the public prototype and its integrated PostgreSQL database.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We used design-thinking to build a practical tool in 7 days:

1. **Empathize & Define:** We learned that workers do not need another open chatbot. They need clear facts, dates, and realistic next steps.
2. **Ideate:** We created the Future Skills Radar (shows which job skills are rising and which skills are declining) and a scenario switch (moderate vs. fast AI growth) so users understand future risks clearly.
3. **Build & Team Roles:** Our 4-person team divided the work cleanly:
   - Darren: Trend data, dates, and scenario switches.

- Keanu: CV skill intake and job-matching scores.
- David: Course catalogue, pathways, SDG mappings, guardrails, 3D element creation, and app deployment.
- Ryan: Frontend UI/UX, page layouts and interactions.

4. **Testing:** Our selected demo set contains five complete synthetic CVs, and CV parsing has been checked against all five. Users confirm extracted skills before saving. Age and gender are excluded from matching inputs, with counterfactual checks supporting review; this does not establish fairness across every population. Recommendations must resolve to catalogue records.

This keeps our project simple, reliable, and ready to show live.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *(e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management.").*	**Main SDG** **SDG 4 (Quality Education):** **Target 4.4:** concerns relevant technical and vocational skills for employment, decent jobs and entrepreneurship. **Adept's contribution:** Identify gaps between users' confirmed skills and target-role requirements, then build prerequisite-aware learning pathways from sourced catalogue courses. Plans account for available study time and budget, while progress tracking and evidence review support continued skills development. **Target 4.3:** concerns equal access to affordable, quality technical, vocational and tertiary education. **Adept's contribution:** Help users compare learning options through visible costs, duration, prerequisites and

Metric Category	Your Response
	source information. Multilingual guidance and personalised planning aim to make relevant training easier to navigate for learners with different resources and circumstances. **Secondary SDGs** SDG 8 (Decent Work and Economic Growth): - Target 8.2: concerns productivity through technological upgrading and innovation - Target 8.5: concerns productive employment, decent work and equal pay. **SDG 5 (Gender Equality):** **Target 5.b:** Use enabling technology, particularly ICT, to support women's empowerment. **Adept's contribution:** Provide women with skills-based career guidance and personalised reskilling pathways. Gender is excluded from matching inputs, while explainable scores help users inspect the basis of recommendations. **SDG 9** (Industry, Innovation and Infrastructure): **Target 9.b:** Support domestic technological development, research and innovation in developing countries. **Adept's contribution:** Support workforce readiness for technological change by connecting AI research signals with relevant skills, future-role matches and structured learning pathways. This is an indirect contribution through workforce capability development. **SDG 10** (Reduced Inequalities): **Target 10.2:** Promote social and economic inclusion irrespective of age, sex, disability, race, ethnicity, origin, religion or economic status. **Adept's contribution:** Support access to reskilling guidance through English, Bahasa Melayu and Chinese interfaces, with pathways that account for users' available study time and budget.

Metric Category	Your Response
	Target 10.3: Promote equal opportunity and reduce inequalities of outcome, including those associated with discriminatory practices. **Adept's contribution:** Use skills-based matching that excludes age, gender and background from scoring inputs. Explainable score components and demographic counterfactual tests support scrutiny of recommendations.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Hays Malaysia reported in March 2025 that 56% of working professionals used AI tools at work, while 26% had received training to use them. These are historical survey findings providing external context. Adept's planning-time baseline will be measured using matched synthetic cases and the same course catalogue.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Catalogue-ID validity rate (%): recommended course IDs that resolve to existing catalogue records divided by all recommended course IDs, multiplied by 100. Also measure median time from starting intake to saving a valid pathway, including prerequisite, time, budget and source checks.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 targets: 100% catalogue-ID validity and median time-to-plan below three minutes across the five synthetic demo profiles. Report task conditions, failures and measured results. These are evaluation targets, not achieved outcomes or a guarantee that every description, price or external link is error-free.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

✅ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

✅ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

✅ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

✅ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: -

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code / dataset / model)	License
Python ≥3.11	code	Python Software Foundation License

Name	Type (code / dataset / model)	License
Node.js 24, pnpm 10.29.3	code	MIT
Next.js, React, React DOM	code	MIT
TypeScript	code	Apache-2.0
Tailwind CSS, @tailwindcss/postcss	code	MIT
next-intl	code	MIT
Three.js (Adi's interactive mascot)	code	MIT
@fontsource-variable/inter	code (font asset)	OFL-1.1 (Inter)
lucide-react	code	ISC
class-variance-authority	code	Apache-2.0
clsx, tailwind-merge	code	MIT
openapi-fetch, openapi-typescript	code	MIT
FastAPI, Pydantic, pydantic-settings	code	MIT
Uvicorn	code	BSD-3-Clause
psycopg 3, psycopg-pool	code	LGPL-3.0-only (binary bundle has additional notices)
python-multipart	code	Apache-2.0
Python standard library	code	PSF-2.0
PyMuPDF	code	AGPL-3.0 or Artifex commercial licence
HTTPX	code	BSD-3-Clause
PostgreSQL 16, pgvector	code	PostgreSQL License
RunPod Python SDK, sentence-transformers, Hugging Face Transformers	code	MIT for RunPod Python SDK; Apache-2.0 for Sentence Transformers and Hugging Face Transformers.

Name	Type (code / dataset / model)	License
PyTorch	code	BSD-3-Clause
pytest, Ruff, ESLint, eslint-config-next	code	MIT
uv	code	MIT or Apache-2.0
Docker Engine and Compose	code	Apache-2.0 (open-source components)
Playwright	code	Apache-2.0
Mermaid	code	MIT
@types/node, @types/react, @types/react-dom, @types/three	code	MIT
GitHub Actions	service	GitHub Terms of Service
sentence-transformers/all-MiniLM-L6-v2 (pinned revision)	model	Apache-2.0
GLM-5.3 (via RunPod)	model	Hosted inference subject to RunPod service terms and applicable model-provider terms.
Google Gemini 3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite (optional fallback)	model	Google Gemini API Terms
Gemini Embedding 001 (optional)	model	Google Gemini API Terms
arXiv cs.AI RSS, query API and cached research windows	dataset / API	arXiv API terms (metadata and paper content have different rights)
U.S. BLS Employment Projections 2023–2033 extract	dataset	U.S. government data; cite BLS
O*NET 31.0 occupation crosswalk	dataset	CC BY 4.0 with stated exceptions (credit and modifications required)
UN SDG Goal and Target APIs and metadata	dataset / API	UN data terms and attribution

Name	Type (code / dataset / model)	License
Public job-posting feeds (Greenhouse, Lever, Ashby, Workday, Jibe, Himalayas, Jobicy, Remote OK, Remotive)	dataset / API	Each board's and employer's own terms; public access is not an open-data licence
Optional keyed job APIs (Careerjet, Jooble, Web3.career)	dataset / API	Provider API terms and credentials required
Job-opening and certification extracts	dataset	Source sites retain rights
Certification and exam-pricing pages (Microsoft, AWS, Google Cloud, IIBA, IAPP)	dataset	Issuer site terms
Microsoft Learn Catalog API	dataset / API	Microsoft API Terms of Use
Coursera course API and pages	dataset / API	Coursera Terms of Use
Other provider course pages (AITraining2U, UNIMAS PROCEL, FMM, MAICSA, UKM, UTM, IBFIM)	dataset	Respective provider copyright and terms
Bank Negara Malaysia exchange-rate API	dataset / API	BNM Terms of Use
Curated skill, course, prerequisite and SDG-alignment mappings; historical trend snapshots	dataset	Team-created; upstream BLS, provider and research rights retained; no UN endorsement implied
Synthetic personas, organisations, regions and CV fixtures	Synthetic personas, organisations, regions, regression CV fixtures and five selected demo CVs	Team-provided synthetic demo and evaluation material; no real identities. Additional intake settings are synthetic demo assumptions.
PyYAML, actionlint, Prettier	code / development tools	MIT
GitHub Container Registry (GHCR)	service	GitHub Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

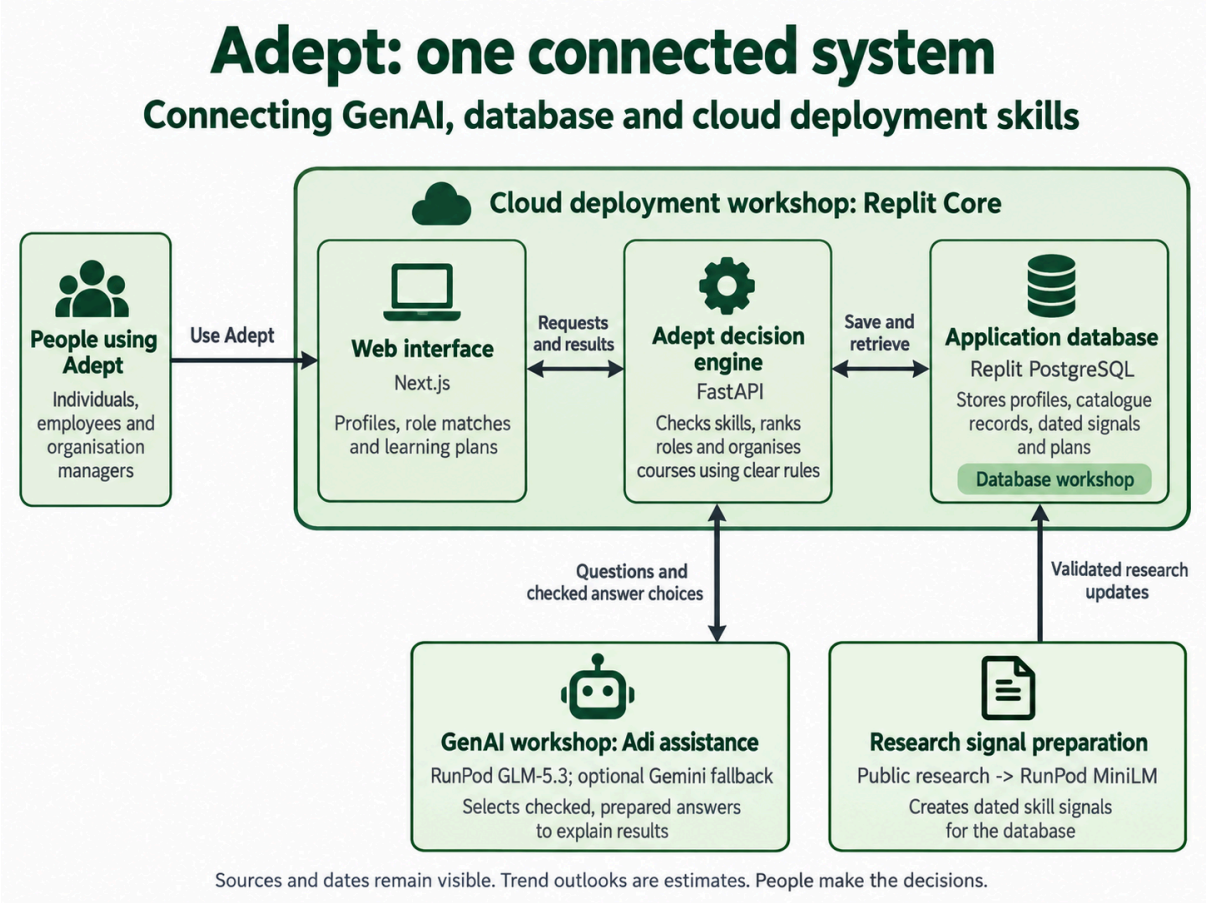
7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Date]: 25/10/2026

Appendix


Adept: one connected system
Connecting GenAI, database and cloud deployment skills
Cloud deployment workshop: Replit Core
People using Adept
Individuals, employees and organisation managers
Use Adept
Web interface
Next.js
Profiles, role matches and learning plans
Requests and results
Adept decision engine
FastAPI
Checks skills, ranks roles and organises courses using clear rules
Save and retrieve
Application database
Replit PostgreSQL
Stores profiles, catalogue records, dated signals and plans
Database workshop
Questions and checked answer choices
Validated research updates
GenAI workshop: Adi assistance
RunPod GLM-5.3; optional Gemini fallback
Selects checked, prepared answers to explain results
Research signal preparation
Public research -> RunPod MiniLM
Creates dated skill signals for the database
Sources and dates remain visible. Trend outlooks are estimates. People make the decisions.